

MODULE – 1

Chapter-1

Foundation of Data

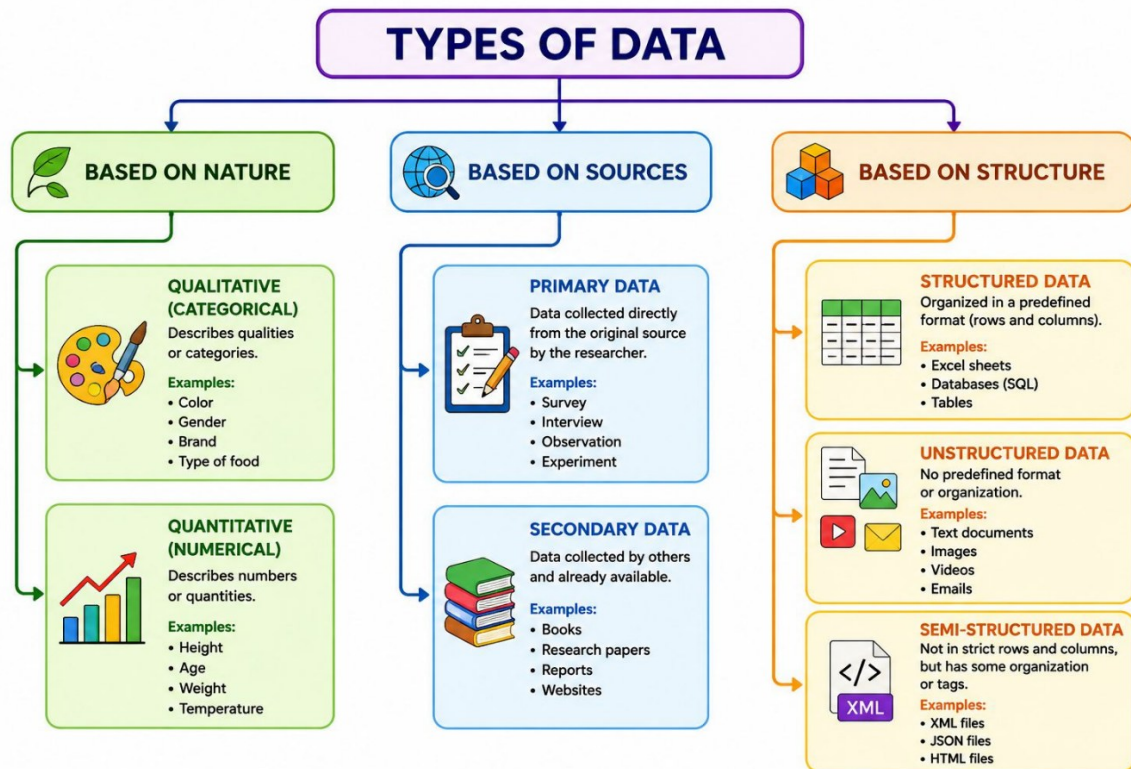
Q. What is Data?

Data means facts, number or information collected for reference or analysis.

Data Surrounds Us



Types of Data Analytics



BASED ON NATURE

Qualitative (Categorical)

➤ Nominal (no. order) –

Data that is Categorical without any order or ranking.

Examples: Gender, Colour, Country, Type of food, Brand.

➤ Ordinal (order) –

Data that has a clear order or ranking, but the gap between them is not defined.

Examples: Rating scale (Poor-Good-Excel) Educational level, Class rank, Satisfaction level.

Quantitative (Numerical)

➤ Discrete – Countable value

Examples: no. of students =25, no. of book, no. of website, no. of shoe size, date & time.

➤ Continuous – Measurable value

Examples: Height, Weight, Temperature, time, speed, amount of water, age, date.

BASED ON SOURCE

➤ **Primary Data**

Collected first hand

Example: survey, experiments, observation, interview

➤ **Secondary Data**

Already available

Example: Reports, Books, Research papers.

BASED ON STRUCTURE

➤ **Structure Data:**

Data that highly organized in rows and columns.

Example: Excel sheet, Databases, Students mark lists.

➤ **Unstructured Data:**

Data that does not have a fixed format.

Example: Text, Images, Videos, Audio.

➤ **Semi-Structured Data:**

Data that has some structure but not fully organized.

Example: Emails, JSON / XML files, Web pages.

CHAPTER-2

DATA ANALYSIS

Data analysis is the process of collecting, cleaning, transforming, and interpreting data to discover meaningful insights, support decision-making and predict future outcomes.

Objectives of Data Analysis

- Identify trends and pattern.
- Support Strategic and operational decisions.
- Convert raw data into actionable knowledge.

Daily life Application

- 1) Online Shopping
- 2) Personal Finance
- 3) Screen Time Analysis
- 4) Instagram Post Analysis
- 5) Fitness Tracking
- 6) Weather Forecast
- 7) Education
- 8) Traffic Management
- 9) Social Media Insights

10)Hospital and Healthcare-work

11)Logistics and Delivery

12)Restaurant Business

13)Agriculture

14)Banking

1. Online Shopping -

Helps websites suggest products based on customer interests and previous purchases.

2. Personal Finance -

Tracks income, expenses, savings, and helps manage budgets better.

3. Screen Time Analysis -

Monitors mobile or app usage to improve productivity and reduce overuse.

4. Instagram Post Analysis -

Analysis likes, comments, shares, and reach to understand audience engagement.

5. Fitness Tracking -

Tracks steps, calories, heart rate, sleep, and workout performance.

6. Weather Forecast -

Uses weather data to predict rain, temperature, and climate conditions.

7. Education -

Helps students and teachers track performance, attendance, and learning progress.

8. Traffic Management -

Analysis road traffic data to reduce congestion and improve transportation.

9. Social Media Insights -

Measures trends, user behaviour, and content performance on social platforms.

10. Hospital and Healthcare Work -

Helps monitor patient records, disease trends, and improves treatment planning.

11. Logistics and Delivery -

Tracks shipments, delivery routes, and delivery time for faster service.

12. Restaurant Business -

Analysis customer preferences, sales, and food demand to improve service.

13. Agriculture -

Helps farmers monitor crop growth, weather, soil quality, and irrigation.

14. Banking -

Detects fraud, manages customer accounts, analyses transactions, and improves security.